

TASK 15: ASSIGNMENT NO. 15

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Domain: Artificial intelligence and machine Learning (AIML)

College Name: Manav School of Polytechnic AKOLA

Branch : Computer Engineering

GitHub Repositories Link: <https://github.com/anujaw193-star/python-project-collection>

Software Required

- Python
- Google Colab
- Pandas
- NumPy

Pandas Commands Used

1. (import pandas as pd)

```
[1] import numpy as np
import pandas as pd
```

2. (pd.read_csv("countries.csv"))

```
*** Choose files Countries.csv
Countries.csv(text/csv) - 95410 bytes, last modified: 04/07/2026 - 100% done
Saving Countries.csv to Countries.csv
User uploaded file "Countries.csv" with lenght 95410 bytes
```

3. (df.head())

```
filepath = '/content/countries.csv'
df = pd.read_csv(filepath)
df.head()
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliament_seats_pct	rural_po
0	Afghanistan	Islamic State of Afghanistan	Afghan afghani	Kabul	Southern Asia	Asia	Afghan	33.0	65.0	383560.0	...	41128771	27.01610	
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.0	20.0	11655.5	...	2775634	35.71430	
2	Algeria	People's Democratic Republic of Algeria	Algerian dinar	Algiers	Northern Africa	Africa	Algerian	28.0	3.0	413588.0	...	44803225	8.10811	
3	Andorra	Principality of Andorra	Euro	Andorra la Vella	Southern Europe	Europe	Andorran	42.5	1.5	187.2	...	79824	46.42860	
4	Angola	People's Republic of Angola	Angolan kwanza	Luanda	Middle Africa	Africa	Angolan	-12.5	18.5	569525.0	...	35588987	33.63640	

5 rows x 64 columns

4. (df.tail())

```
[6]
✓ 0s # Display last 5 rows
print("\nLast 5 Rows:")
print(df.tail())
```

Output:

```
Last 5 Rows:
... 189      Vietnam Socialist Republic of Vietnam Vietnamese dong \
190 West Bank and Gaza West Bank and Gaza Israeli new shekel
191      Yemen Republic of Yemen Yemeni rial
192      Zambia Republic of Zambia New Zambian kwacha
193      Zimbabwe Republic of Zimbabwe U.S. dollar

      capital_city      region continent      demonym      latitude \
189      Hanoi South-Eastern Asia Asia Vietnamese 16.166667
190      Ramallah Western Asia Asia Palestinian 31.900000
191      Sana'a Western Asia Asia Yemeni 15.000000
192      Lusaka Eastern Africa Africa Zambian -15.000000
193      Harare Eastern Africa Africa Zimbabwean -20.000000

      longitude agricultural_land ... population \
189 107.833333 123600.0 ... 98186856
190 35.200000 4449.0 ... 5043612
191 48.000000 234520.0 ... 33696614
192 30.000000 238360.0 ... 20017675
193 30.000000 162000.0 ... 16320537

      women_parliament_seats_pct rural_population urban_population press \
189 30.2605 60123739 38063117 1.04
190 NaN 1145354 3898258 1.75
191 0.0000 20491585 13205029 1.27
192 15.0602 10857387 9160288 2.15
193 30.5660 11033499 5287038 1.85

      democracy_score democracy_type median_age political_leader \
189 3.08 Authoritarian 28.4 Nguyễn Phú Trọng
190 4.39 Hybrid regime 15.4 NaN
191 1.95 Authoritarian 14.8 Abdrabbuh Mansur Hadi
192 5.61 Hybrid regime 13.2 Edgar Lungu
193 3.16 Authoritarian 14.2 Emmerson Mnangagwa
```

5. (df.shape)

Code:

```
[7]
✓ 0s # Display shape
print("\nShape:")
print(df.shape)
```

Output:

```
Shape:
(194, 64)
```

6. (df.columns)

Code:

```
[8]
✓ 0s # Column names
print("/Column Names:")
print(df.columns)
```

Output:

```

/Column Names:
Index(['country', 'country_long', 'currency', 'capital_city', 'region',
      'continent', 'demonym', 'latitude', 'longitude', 'agricultural_land',
      'forest_area', 'land_area', 'rural_land', 'urban_land',
      'central_government_debt_pct_gdp', 'expense_pct_gdp', 'gdp',
      'inflation', 'self_employed_pct', 'tax_revenue_pct_gdp',
      'unemployment_pct', 'vulnerable_employment_pct',
      'electricity_access_pct', 'alternative_nuclear_energy_pct',
      'electricity_production_coal_pct',
      'electricity_production_hydroelectric_pct',
      'electricity_production_gas_pct', 'electricity_production_nuclear_pct',
      'electricity_production_oil_pct', 'electricity_production_renewable_pct',
      'energy_imports_pct', 'fossil_energy_consumption_pct',
      'renewable_energy_consumption_pct', 'co2_emissions',
      'methane_emissions', 'nitrous_oxide_emissions',
      'greenhouse_other_emissions', 'urban_population_under_5m',
      'health_expenditure_pct_gdp', 'health_expenditure_capita',
      'hospital_beds', 'hiv_incidence', 'suicide_rate', 'armed_forces',
      'internally_displaced_persons', 'military_expenditure_pct_gdp',
      'birth_rate', 'death_rate', 'fertility_rate', 'internet_pct',
      'life_expectancy', 'net_migration', 'population_female',
      'population_male', 'population', 'women_parliament_seats_pct',
      'rural_population', 'urban_population', 'press', 'democracy_score',
      'democracy_type', 'median_age', 'political_leader', 'title'],
      dtype='object')

```

7. (df.info())

Code:

```

[9] # Dataset information
✓ Os print("\nDataset Information:")
print(df.info())

```

Output:

```

...
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 194 entries, 0 to 193
Data columns (total 64 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   country                                   194 non-null    object
1   country_long                             194 non-null    object
2   currency                                 194 non-null    object
3   capital_city                             194 non-null    object
4   region                                  194 non-null    object
5   continent                                194 non-null    object
6   demonym                                  194 non-null    object
7   latitude                                 194 non-null    float64
8   longitude                                194 non-null    float64
9   agricultural_land                       193 non-null    float64
10  forest_area                             194 non-null    float64
11  land_area                               194 non-null    float64
12  rural_land                              194 non-null    float64
13  urban_land                              194 non-null    float64
14  central_government_debt_pct_gdp         120 non-null    float64
15  expense_pct_gdp                         156 non-null    float64
16  gdp                                       193 non-null    float64
17  inflation                                184 non-null    float64
18  self_employed_pct                       179 non-null    float64
19  tax_revenue_pct_gdp                     159 non-null    float64
20  unemployment_pct                         179 non-null    float64
21  vulnerable_employment_pct               179 non-null    float64
22  electricity_access_pct                   194 non-null    float64

22  electricity_access_pct                   194 non-null    float64
23  alternative_nuclear_energy_pct          138 non-null    float64
24  electricity_production_coal_pct          138 non-null    float64
25  electricity_production_hydroelectric_pct 138 non-null    float64
26  electricity_production_gas_pct           138 non-null    float64
27  electricity_production_nuclear_pct       138 non-null    float64
28  electricity_production_oil_pct           138 non-null    float64
29  electricity_production_renewable_pct      138 non-null    float64
30  energy_imports_pct                      138 non-null    float64
31  fossil_energy_consumption_pct            168 non-null    float64
32  renewable_energy_consumption_pct         192 non-null    float64
33  co2_emissions                           191 non-null    float64
34  methane_emissions                       191 non-null    float64
35  nitrous_oxide_emissions                 191 non-null    float64
36  greenhouse_other_emissions              184 non-null    float64
37  urban_population_under_5m                194 non-null    float64
38  health_expenditure_pct_gdp              190 non-null    float64
39  health_expenditure_capita               190 non-null    float64
40  hospital_beds                           192 non-null    float64
41  hiv_incidence                           133 non-null    float64
42  suicide_rate                            183 non-null    float64
43  armed_forces                             174 non-null    float64
44  internally_displaced_persons              73 non-null    float64
45  military_expenditure_pct_gdp            163 non-null    float64
46  birth_rate                              194 non-null    float64
47  death_rate                              194 non-null    float64
48  fertility_rate                           193 non-null    float64
49  internet_pct                             194 non-null    float64
50  life_expectancy                          191 non-null    float64
51  net_migration                           194 non-null    int64
52  population_female                        194 non-null    int64
53  population_male                          194 non-null    int64
54  population                               194 non-null    int64
55  women_parliament_seats_pct              193 non-null    float64

```

```

54 population 194 non-null int64
55 women_parliament_seats_pct 193 non-null float64
56 rural_population 194 non-null int64
57 urban_population 194 non-null int64
58 press 194 non-null float64
59 democracy_score 194 non-null float64
60 democracy_type 194 non-null object
61 median_age 194 non-null float64
62 political_leader 187 non-null object
63 title 187 non-null object
dtypes: float64(48), int64(6), object(10)
memory usage: 97.1+ KB
None

```

8. (df.describe())

Code:

```

[17]
✓ Os # Describe
print(df.describe())

```

Output:

```

**
count    latitude    longitude    agricultural_land    forest_area    land_area  \
mean      18.975601    22.027491    2.454551e+05    2.086784e+05    6.675087e+05
std       23.876225    66.396389    6.356268e+05    7.824926e+05    1.837107e+06
min      -41.000000   -175.000000    4.000000e+00    0.000000e+00    2.027000e+00
25%       4.000000    -5.000000    6.464000e+03    3.331775e+03    2.355250e+04
50%      16.583333    21.500000    3.872780e+04    2.528925e+04    1.203750e+05
75%      40.000000    50.162500    2.150000e+05    1.236735e+05    5.237000e+05
max       65.000000   178.000000    5.285080e+06    8.153120e+06    1.637690e+07

count    rural_land    urban_land    central_government_debt_pct_gdp  \
mean    6.563711e+05    9777.116531    66.759366
std     1.811169e+06    42301.458421    71.806247
min     3.495450e-02     0.000000     0.000000
25%     2.186562e+04    359.618250    31.951300
50%     1.159945e+05    1645.170000    55.426850
75%     4.911505e+05    4054.022500    79.539300
max     1.622420e+07    522345.000000    687.994000

count    expense_pct_gdp    gdp    ...    net_migration    population_female  \
mean      30.051403    5.144851e+11    ...      -51.407216    2.028316e+07
std       26.740880    2.307148e+12    ...     94525.968598    7.258941e+07
min        0.000267    6.034940e+07    ...   -525116.000000    5.513000e+03
25%       18.371875    1.181390e+10    ...   -12242.250000    1.036218e+06
50%       27.337750    4.115390e+10    ...    -970.000000    4.502713e+06
75%       35.083500    2.519450e+11    ...     2904.250000    1.526606e+07
max       310.443000    2.546270e+13    ...    561580.000000    6.915285e+08

```

75%	35.083500	2.519450e+11	...	2904.250000	1.5				
max	310.443000	2.546270e+13	...	561580.000000	6.9				
	population_male	population	women_parliament_seats_pct						
count	1.940000e+02	1.940000e+02	193.000000						
mean	2.050572e+07	4.078888e+07	25.022994						
std	7.607158e+07	1.486470e+08	12.671044						
min	5.799000e+03	1.131200e+04	0.000000						
25%	1.044902e+06	2.106358e+06	15.384600						
50%	4.450049e+06	9.125614e+06	25.252500						
75%	1.478872e+07	3.031361e+07	33.636400						
max	7.311805e+08	1.417173e+09	61.250000						
	rural_population	urban_population	press	democracy_s					
count	1.940000e+02	1.940000e+02	194.000000	194.00					
mean	1.763322e+07	2.315566e+07	2.539330	4.64					
std	7.664187e+07	7.940393e+07	1.800128	2.81					
min	0.000000e+00	5.717000e+03	0.000000	0.00					
25%	5.896640e+05	1.222244e+06	1.525000	2.72					
50%	2.512382e+06	4.508837e+06	2.400000	5.05					
75%	1.133354e+07	1.621355e+07	2.925000	6.96					
max	9.088048e+08	8.975784e+08	10.000000	9.87					
	median_age								
count	194.000000								
mean	25.661856								
...									
Terminal									
std	7.664187e+07	7.940393e+07	1.800128	2.818297					
min	0.000000e+00	5.717000e+03	0.000000	0.000000					
25%	5.896640e+05	1.222244e+06	1.525000	2.722500					
50%	2.512382e+06	4.508837e+06	2.400000	5.050000					
75%	1.133354e+07	1.621355e+07	2.925000	6.967500					
max	9.088048e+08	8.975784e+08	10.000000	9.870000					
	median_age								
count	194.000000								
mean	25.661856								
std	9.415569								
min	10.500000								
25%	16.950000								
50%	24.950000								
75%	34.050000								
max	50.500000								
[8 rows x 54 columns]									

9. (df.dtypes)

Code:

```
[10]
✓ 0s # Data types
print("\nData Types:")
print(df.dtypes)
```

Output:

```

Data Types:
country          object
country_long     object
currency         object
capital_city     object
region          object
...
democracy_score  float64
democracy_type   object
median_age       float64
political_leader  object
title           object
Length: 64, dtype: object

```

10. (df.isnull().sum())

Code:

```

[11] # Statistical summary
✓ 0s # print("\nStatistical Summary:")
    print(df.describe())

```

Output:

```

std      23.876225   66.396389   6.356268e+05   7.824926e+05   1.837107e+06
min      -41.000000  -175.000000   4.000000e+00   0.000000e+00   2.027000e+00
25%       4.000000   -5.000000   6.464000e+03   3.331775e+03   2.355250e+04
50%      16.583333   21.500000   3.872780e+04   2.528025e+04   1.203750e+05
75%      40.000000   50.162500   2.150000e+05   1.236735e+05   5.237000e+05
max      65.000000  178.000000   5.285080e+06   8.153120e+06   1.637690e+07

      rural land   urban land  central_government_debt_pct_gdp \
count  1.940000e+02   194.000000   120.000000
mean   6.563711e+05   9777.116531   66.759366
std    1.811160e+06   42301.458421   71.806247
min    3.495450e-02   0.000000   0.000000
25%    2.186562e+04   359.618250   31.951300
50%    1.159945e+05   1645.170000   55.426850
75%    4.911505e+05   4054.022500   79.539300
max    1.622420e+07   522345.000000   687.994000

      expense_pct_gdp   gdp   ... net_migration  population_female \
count      156.000000  1.930000e+02   ...   194.000000   1.940000e+02
mean       30.051403  5.144851e+11   ...   -51.407216   2.028316e+07
std       26.740880  2.307148e+12   ...   94525.968598   7.258941e+07
min        0.000267  6.034940e+07   ...  -525116.000000   5.513000e+03
25%       18.371875  1.181390e+10   ...  -12242.250000   1.036218e+06
50%       27.337750  4.115390e+10   ...   -970.000000   4.502713e+06
75%       35.083500  2.519450e+11   ...   2904.250000   1.526606e+07
max       310.443000  2.546270e+13   ...  561580.000000   6.915285e+08

      expense_pct_gdp   gdp   ... net_migration  population_female \
count      156.000000  1.930000e+02   ...   194.000000   1.940000e+02
mean       30.051403  5.144851e+11   ...   -51.407216   2.028316e+07
std       26.740880  2.307148e+12   ...   94525.968598   7.258941e+07
min        0.000267  6.034940e+07   ...  -525116.000000   5.513000e+03
25%       18.371875  1.181390e+10   ...  -12242.250000   1.036218e+06
50%       27.337750  4.115390e+10   ...   -970.000000   4.502713e+06
75%       35.083500  2.519450e+11   ...   2904.250000   1.526606e+07
max       310.443000  2.546270e+13   ...  561580.000000   6.915285e+08

      population_male  population  women_parliament_seats_pct \
count      1.940000e+02  1.940000e+02   193.000000
mean       2.050572e+07  4.078888e+07   25.022994
std       7.607158e+07  1.486470e+08   12.671044
min       5.799000e+03  1.131200e+04   0.000000
25%       1.044902e+06  2.106358e+06   15.384600
50%       4.450049e+06  9.125614e+06   25.252500
75%       1.478872e+07  3.031361e+07   33.636400
max       7.311805e+08  1.417173e+09   61.250000

```

```

rural_population  urban_population  press  democracy_score  \
count    1.940000e+02    1.940000e+02  194.000000    194.000000
mean     1.763322e+07    2.315566e+07   2.539330    4.644536
std      7.664187e+07    7.940393e+07   1.800128    2.818297
min      0.000000e+00    5.717000e+03   0.000000    0.000000
25%     5.896640e+05    1.222244e+06   1.525000    2.722500
50%     2.512382e+06    4.508837e+06   2.400000    5.050000
75%     1.133354e+07    1.621355e+07   2.925000    6.967500
max      9.088048e+08    8.975784e+08  10.000000    9.870000

count    median_age
mean     25.661856
std      9.415569
min      10.500000
25%     16.950000
50%     24.950000
75%     34.050000
max      50.500000

[8 rows x 54 columns]

```

11. (nunique ())

Code:

```

[13] # Unique values count
print("\nUnique Values Count:")
print(df.nunique())

```

Output:

```

...
Unique Values Count:
country          194
country_long     194
currency         142
capital_city     194
region           22
...
democracy_score  146
democracy_type   5
median_age      143
political_leader 187
title            17
Length: 64, dtype: int64

```

12. (df.sample (5))

Code:

```

[14] # Random sample rows
print("\nRandom Sample Rows:")
print(df.sample(5))

```

Output :

Random Sample Rows:							
	country		country_long		currency	\	
136	Poland		Republic of Poland		Polish zloty		
46	Denmark		Kingdom of Denmark		Danish krone		
188	Venezuela	República Bolivariana de Venezuela	Venezuela		bolivar fuerte		
64	Ghana		Republic of Ghana		New Ghanaian cedi		
6	Argentina		Argentine Republic		Argentine peso		
	capital_city	region	continent	demonym	latitude	longitude	\
136	Warsaw	Eastern Europe	Europe	Polish	52.0	20.0	
46	Copenhagen	Northern Europe	Europe	Danish	56.0	10.0	
188	Caracas	South America	Americas	Venezuelan	8.0	-66.0	
64	Accra	Western Africa	Africa	Ghanaian	8.0	-2.0	
6	Buenos Aires	South America	Americas	Argentine	-34.0	-64.0	
	agricultural_land	...	population	women_parliament_seats_pct	\		
136	144610.0	...	37561599	28.2609			
46	26199.9	...	5903037	43.5754			
188	215000.0	...	28301696	22.1557			
64	126037.0	...	33475870	14.5455			
6	1083820.0	...	46234830	44.7471			
	rural_population	urban_population	press	democracy_score	\		
136	14974307	22587292	2.71	6.67			
46	686700	5216337	7.92	9.22			
188	3289223	25012473	1.59	3.16			
64	13853989	19621881	3.76	6.63			
6	3538352	42696478	2.76	7.02			
	democracy_type	median_age	political_leader	title			
136	Flawed democracy	36.3	Mateusz Morawiecki	Prime Minister			
46	Full democracy	37.2	Mette Frederiksen	Prime Minister			
188	Authoritarian	24.2	Nicolás Maduro	President			
64	Flawed democracy	16.6	Nana Akufo-Addo	President			
6	Flawed democracy	27.6	Alberto Fernández	President			

[5 rows x 64 columns]

13. (df.value_counts())

Code:

```
[16]
✓ Os # Value counts of a selected column
print("\nRegion Value Counts:")
print(df['region'].value_counts())
```

Output:

```
Region Value Counts:
region
Eastern Africa      17
Western Asia        17
Western Africa      16
Southern Europe     15
Caribbean          13
South America       12
South-Eastern Asia  11
Northern Europe     10
Middle Africa       10
Eastern Europe      10
Western Europe       9
Southern Asia        9
Central America      8
Northern Africa      6
Micronesia           5
Southern Africa      5
Eastern Asia         5
Central Asia         5
Melanesia            4
Polynesia            3
Australia and New Zealand 2
Northern America     2
Name: count, dtype: int64
```